

WISL -- Conditional Approval to Proceed

Mission-memory layer with 3D failure visualisation and edge-native SLM log normalisation . Neo-Battalion . 3-month programme

To: Sua Isaac; Inessa Wong Manting; Sylvester Lim Huan Kiat (Team WISL)
From: Gary Ang, NUS Enterprise / DVL
Re: WISL proposal -- review outcome and steers for the build

The idea is distinctive -- it is the one proposal in the cohort with no real overlap -- and the engineering instincts are mature: edge-first, no-cloud, low-overhead rendering, and grammar-constrained decoding to keep the SLM honest. That is a good foundation, and the team is well-matched to it: DSO operational access, time-series data expertise, and agentic infrastructure engineering. The thing worth wrestling with is not whether you can build it -- it is how big the impact really is, and where it concentrates. We are glad to approve you to proceed; the points below are offered as steers to fold into your kickoff plan, not hurdles to clear.

What is working

The thesis is right: robotic failures are valuable data, and today that data is lost -- trapped in binary SD-card formats, scattered across disconnected silos, inaccessible without an engineer in the loop. The architecture is well-reasoned. The two-mode ingestion design is smart and not obvious (burst-before-loss for wireless assets, automated cloud-sync for fibre-optic), the edge-first stack is the right call for contested comms, and the grammar-constrained decoding to keep the SLM honest is a mature instinct for a safety-relevant pipeline. The closing line -- 'transient mission failures into permanent institutional assets' -- is the truest sentence in the proposal. It is worth building around.

The thing worth being clear-eyed about

It is worth being clear-eyed about what kind of capability this is. WISL is an enabler -- it improves the learning loop; it does not itself kill a drone, win an engagement, or defend a site. That is real value, but indirect, and indirect value is easy to under-realise and easy for a larger platform to absorb as a feature. The honest win is 'the next mission is better informed,' rather than a direct battlefield effect -- and that smaller claim tends to be the more defensible one.

In the same spirit, the current impact framing may be optimistic in a few ways. Attributable drones are designed to be lost -- you would not run forensics on every expended low-cost FPV, so the payoff sits on the smaller subset of assets worth diagnosing. Most mass-drone losses are not especially mysterious either (shot down, jammed, flat battery), so the 'rare transient condition needing reconstruction' is a thinner slice than it first appears. And 'tactical adjustment within minutes' may be the wrong loop to aim at: in a fast fight the asset is already gone, and the value more likely lives in fleet- and doctrine-level learning over days and weeks. None of this diminishes the idea -- it just points to where to aim it.

Steers worth folding in

1. Consider reframing the impact as an enabler. The strongest version of the claim is that WISL turns transient failures into a permanent, queryable institutional record that makes the next mission better informed -- rather than something that runs forensics on every failure or adjusts

tactics in real time. The smaller, truer claim is the more durable one.

2. Aim at where the impact most likely concentrates. Two cases look like they pay off: expensive, reusable platforms (ISR drones, UGVs, loyal-wingman class) where each loss is costly and diagnosable; and systematic, fleet-wide failure modes -- a batch defect, a firmware bug, a common EW vulnerability -- caught once and propagated across the fleet. That second case may be the highest-leverage one, and it is your own institutional-assets line; it could be worth leading with. The Singapore angle reinforces this: the SAF operates fewer, costlier, reusable systems -- the regime where forensics tends to pay -- rather than mass swarms in dense urban.
3. It may help to anchor the demo to fleet- and doctrine-level learning rather than real-time forensics. By week 12, a compelling proof would be the engineer-free loop on a real failure log -- ingest a PX4 or ArduPilot log automatically, reconstruct the failure path in 3D, and surface a plain-text root-cause insight a non-engineer operator can read -- followed by the fleet payoff: a systematic vulnerability spotted across a small corpus of logs and the fix propagated. 'All three components run' is a weaker story than 'failure log to fleet-wide fix, no specialist in the loop.'
4. The vendor-agnostic normalisation layer is probably where the real defensibility sits, so it may be worth treating as the core rather than the renderer. Value tends to accrue to whoever owns the fleet/data layer, and a standalone forensic tool risks becoming a feature a drone-OS vendor (Auterion, Palantir) bolts on. Turning messy, multi-vendor failure data into one queryable record the operator owns -- without vendor permission -- is the part that could be a product rather than a feature. Validating that layer early would help: in weeks 1-2, Isaac's DSO access could surface real logs from a couple of vendors, since synthetic data will not reveal the ULog, ArduPilot and hexadecimal idiosyncrasies that are the whole point of normalisation.

WISL is credible and execution-ready. The engineering is sound and the architecture is genuinely differentiated -- the work over these 12 weeks is mostly to aim it well: at the assets worth diagnosing, the fleet-wide failure modes, and the normalisation layer that makes it a thing rather than a feature. Fold these steers in where they help, and you are cleared to start.

It need not win the engagement to be valuable -- making the next one better informed is enough, and owning the record no vendor can take from you is what makes it last.

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